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EX NO: 10

**CALCULATION OF PROCESSING GAIN AND GENERATION OF PN SEQUENCES FOR SPREAD SPECTRUM
COMMUNICATION USING MATLAB SIMULATION**

OBJECTIVES

- i) To calculate the processing gain for various PN sequence lengths and tabulate the corresponding processing gain values.
 - ii) To visualize and compare the processing gain obtained for different PN sequence lengths using MATLAB plots and analyze the effect of sequence length on spread spectrum performance.
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PROGRAM (MATLAB CODE)

```
clc;

clear;

close all;

% PN sequence lengths
PN_length = [7 15 31 63 127 255 511 1023];

% Processing Gain (Linear)
PG_linear = PN_length;

% Processing Gain in dB
PG_dB = 10*log10(PG_linear);

% Display table
T = table(PN_length',PG_dB',...
    'VariableNames',{'PN_Length','Processing_Gain_dB'});

disp('Processing Gain Table');

disp(T);

% Plot
figure;

stem(PN_length,PG_dB,'filled');

grid on;
```

```
xlabel('PN Sequence Length');  
ylabel('Processing Gain (dB)');  
title('Spread Spectrum: Processing Gain vs PN Sequence Length');
```

PROCEDURE

Objective (i): Processing Gain Calculation

1. Open MATLAB and create a new script file.
2. Initialize the MATLAB workspace using `clc`, `clear`, and `close all`.
3. Define the PN sequence lengths as: **[7 15 31 63 127 255 511 1023]**.
4. Calculate the processing gain in linear scale (equal to PN sequence length).
5. Convert the processing gain into decibels using the formula:
 $PG(dB) = 10 \log_{10}(PN \text{ length})$.
6. Create a table containing PN sequence lengths and corresponding processing gain values.
7. Display the table in the MATLAB command window.
8. Observe the variation in processing gain for different PN sequence lengths.

Result:

The processing gain values for different PN sequence lengths were calculated successfully. It was observed that the processing gain increases as the PN sequence length increases.

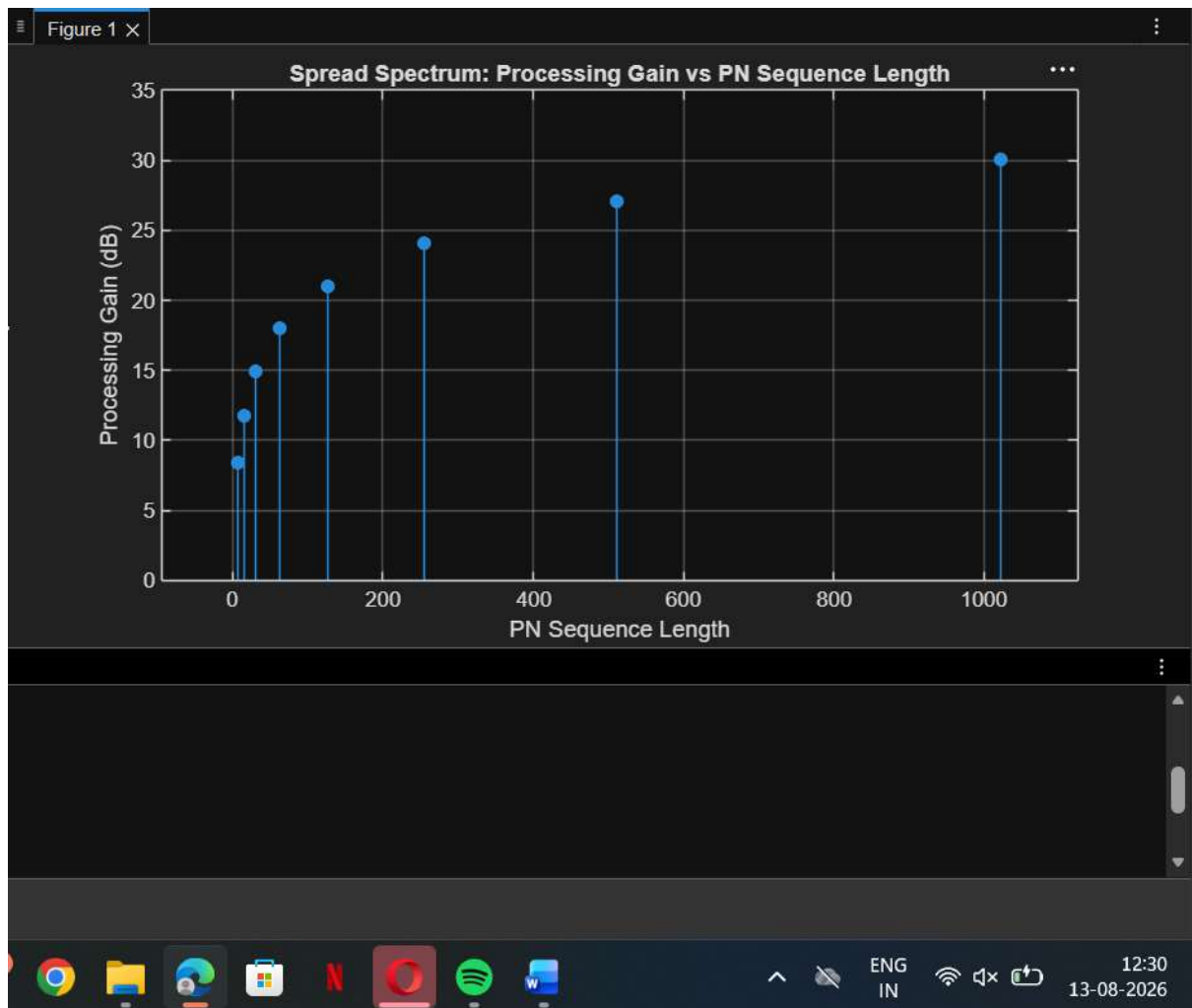
Objective (ii): Visualization Using MATLAB Plot

1. Create a figure window in MATLAB.
2. Plot PN sequence length on the x-axis.
3. Plot processing gain (in dB) on the y-axis.
4. Use the `stem()` function to display discrete values.
5. Label the axes appropriately.
6. Provide a suitable title to the plot.
7. Enable the grid for better visualization.
8. Compare processing gain values for different PN lengths and observe the trend.

Result:

The MATLAB plot successfully illustrated the relationship between PN sequence length and processing gain. Longer PN sequences produced higher processing gains, thereby improving interference rejection and communication reliability.

OUTPUT (GRAPH)



FINAL RESULT

The processing gain for various PN sequence lengths was successfully calculated and tabulated using MATLAB. The graphical analysis showed that processing gain increases with PN sequence length, confirming that longer sequences enhance spread spectrum system performance by improving noise immunity and interference rejection.